



Intellectual Property Protection Document

GreenhouseOS 5.0 — Enterprise Virtual Twin Platform

PREPARED BY	Growa
DOCUMENT TYPE	Technical IP Reference
VERSION	5.0.0
DATE	September 2026
APPROACH	Black-box architecture

Growa IP Protection Vault

Classification · Confidential IP Reference

Product · GreenhouseOS 5.0 by Growa

Approach · Black-box architecture — interfaces, outcomes, and value only

Navigation

Note	Focus
00-Cover	Title, legal notice, design charter
01-Platform-Value	Strategic value pillars and differentiation
02-Black-Box-Architecture	System topology without internal algorithms
03-Agronomic-Intelligence	Agronomic engine capabilities (abstracted)
04-Climate-Twin	Real-time climate twin (abstracted)
05-Virtual-Experience	3D design experience (abstracted)
06-GAIA-Intelligence	GAIA copilot layer (abstracted)
07-Data-Governance	Persistence, auth, and access control
08-Industrial-Bridge	Climate-computer interoperability
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Design Principles

This vault communicates **what the platform delivers** and **how subsystems connect**, never **how proprietary algorithms are implemented**.

OBSERVABLE LAYER	→ Documented in this vault
INTERFACE LAYER	→ Sanitized contracts only
PROPRIETARY CORE	→ Black box – not disclosed

Identity	
Field	Value
Company	Growa
Platform	GreenhouseOS 5.0
Category	Enterprise 3D Virtual Twin & Agronomic SaaS
License	BUSL-1.1 (Change Date 2029-08-03)

See *10-Trade-Secret-Policy* before distribution.

Platform Value

Home · 02-Black-Box-Architecture

Vision

GreenhouseOS by **Growa** transforms greenhouse design from static CAD into a **living virtual twin** — every design decision immediately surfaces financial, agronomic, and engineering consequences.

Four Value Pillars

Energy & Economics Real-time OPEX modelling, energy intensity (kWh/m²), and ROI scenarios for climate equipment investments. ### Agronomic Intelligence Industry-standard microclimate indices — vapour pressure deficit, daily light integral, and crop stress signals — computed from design context. ### Engineering Outputs Actionable sizing guidance: heating capacity, dehumidification demand, and ventilation exchange rates. ### Industrial Bridge Export of climate-control setpoints compatible with leading greenhouse automation vendors.

What Makes Growa Different

Capability	Outcome
No-Form Design	Edit greenhouse geometry directly in 3D — no disconnected forms
Unified Twin	One model drives visuals, physics preview, and export
GAIA Copilot	Natural-language climate guidance powered by Growa
Deterministic Fallback	Platform remains operational without external AI dependency
Multilingual	English, Italian, Spanish, French

Maturity

Six production milestones delivered: physics core, 3D canvas, real-time simulation, interactive shaders, AI gateway, and enterprise data layer.

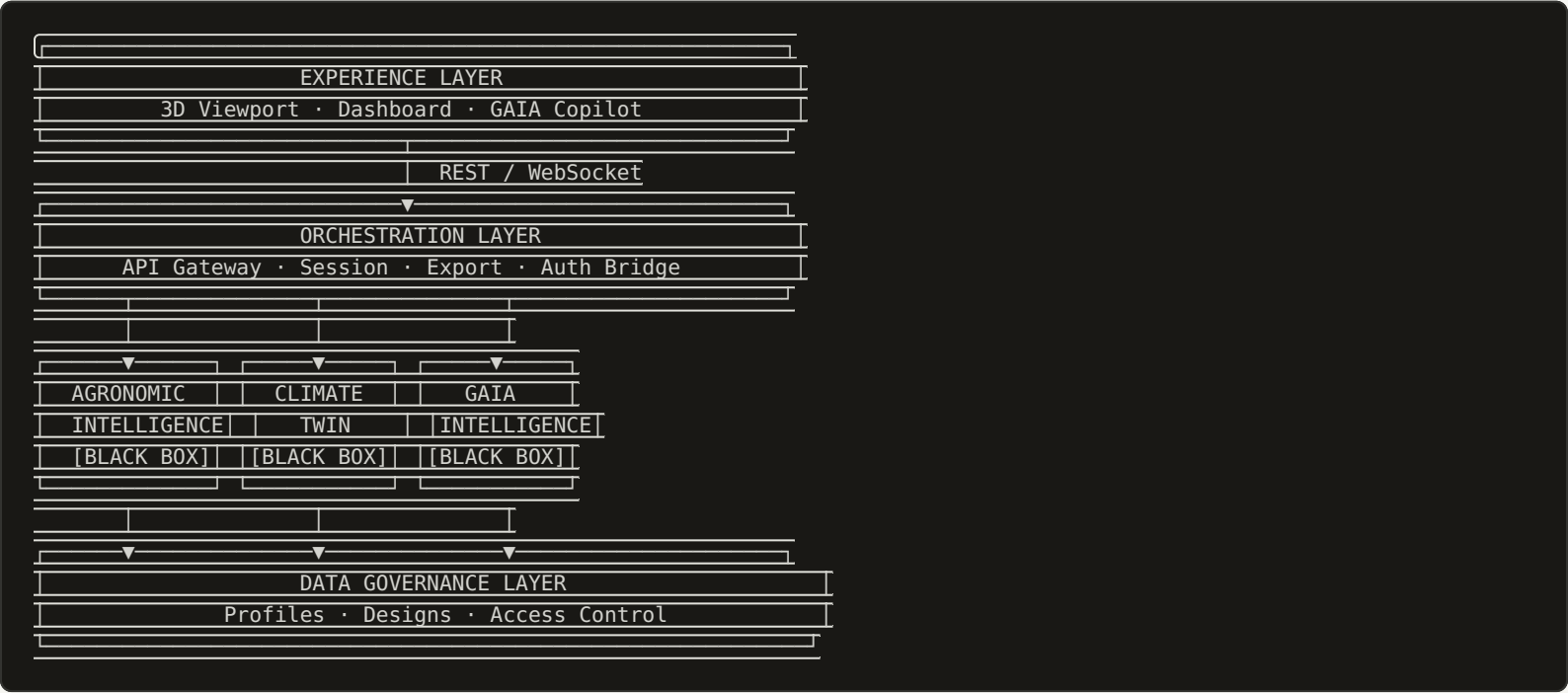
Continue to 02-Black-Box-Architecture

Black-Box Architecture

Home · 01-Platform-Value · 03-Agronomic-Intelligence

System Topology

High-level view of the Growa platform. Internal algorithmic cores are intentionally opaque.



Layer Responsibilities

Layer	Role	Disclosed
Experience	User interaction, visualization, localization	Behaviour only
Orchestration	Routing, validation, export formatting	API contracts
Agronomic Intelligence	Crop microclimate indices	Inputs/outputs
Climate Twin	Real-time energy & equipment simulation	Capabilities
GAIA Intelligence	AI-assisted climate guidance	Interface
Data Governance	Persistence, auth, row-level security	Schema shape

Sanitized Platform Contract

The entire platform is addressable through a single design context object. Implementation is proprietary.

```
/** Public design context – sanitized interface */
interface GrowaDesignContext {
  geometry: GreenhouseGeometry;
  covering: CoveringMaterial;
  crop: CropProfile;
  location: GeoCoordinate;
  equipment?: ClimateEquipmentProfile;
}

/** Platform orchestrator – black-box entry point */
interface GrowaPlatform {
  previewMicroclimate(ctx: GrowaDesignContext): MicroclimateSnapshot;
  streamSimulation(ctx: GrowaDesignContext): AsyncStream<SimulationFrame>;
  exportSetpoints(ctx: GrowaDesignContext, vendor: VendorFormat): SetpointBundle;
  askGAIA(ctx: GrowaDesignContext, query: string): GuidanceResponse;
}
```

Note: Type names illustrate structure only. Proprietary solvers, coefficients, and calibration data are not disclosed.

Technology Stack (Non-Secret)

Tier	Stack
Backend	Python · FastAPI · Pydantic
Frontend	React · TypeScript · Three.js
Real-time	WebSocket
Data	PostgreSQL with row-level security
AI	Multi-provider gateway with local fallback

Continue to [03-Agronomic-Intelligence](#)

Agronomic Intelligence

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Capability Summary

The **Agronomic Intelligence** module computes crop-relevant microclimate indices from meteorological inputs and greenhouse design parameters. It adheres to internationally recognised agronomic standards without exposing proprietary calibration logic.

Observable Outputs

Output	Unit	Purpose
Reference evapotranspiration	mm/day	Water demand baseline
Vapour pressure deficit	kPa	Stomatal / transpiration driver
Daily light integral	mol/m ² /day	Photosynthesis adequacy
Stress indices	0–1	Crop-stage deviation signals

Black-Box Interface

```
/** Agronomic engine – implementation is proprietary */
interface AgronomicEngine {
  evaluate(context: AgronomicInput): AgronomicSnapshot;
}

interface AgronomicInput {
  climate: ExternalClimate;
  covering: CoveringMaterial;
  crop: CropProfile;
}

interface AgronomicSnapshot {
  et0: number; // mm/day
  vpdKpa: number;
  dliMolM2Day: number;
  stressIndices: StressMap;
}
```

Standards Alignment (Public Knowledge)

The engine is **aligned with** — but not a reproduction of — published references:

- FAO Irrigation and Drainage Paper 56 (reference evapotranspiration)
- ASHRAE psychrometric fundamentals (humidity calculations)

Supported Crop Families

Tomato · Cucumber · Pepper · Lettuce · Strawberry · Cannabis

Each crop supports multiple growth stages with stage-aware optimal bands. Specific coefficient tables are proprietary.

Continue to [04-Climate-Twin](#)

Climate Twin

Capability Summary

The **Climate Twin** module provides real-time greenhouse energy and microclimate simulation. It couples agronomic transpiration with envelope physics and climate-equipment models to produce internal temperature, humidity, and spatial heatmaps.

Observable Outputs

Output	Description
Thermal balance	Solar, transpiration, ventilation, conduction fluxes
Internal microclimate	Temperature, RH, VPD at steady state
Spatial heatmap	Per-zone temperature and VPD field
Equipment loads	Heating, cooling, fan energy demand

Black-Box Interface

```
/** Climate twin – proprietary solver inside */
interface ClimateTwin {
  solve(context: TwinInput): TwinSnapshot;
  generateHeatmap(context: TwinInput): HeatmapField;
}

interface TwinInput extends GrowaDesignContext {
  externalClimate: ExternalClimate;
}

interface TwinSnapshot {
  internalTempC: number;
  internalRhPct: number;
  vpdKpa: number;
  energyFluxes: EnergyBalance;
}
```

Equipment Categories (Abstracted)

Category	Modelled Behaviour
Ventilation	Mechanical, natural, and infiltration exchange
Evaporative cooling	Pad walls, fog systems
Mechanical AC	Rated cooling with duct distribution
Heating	Localised warm zones
Circulation	Air mixing without direct exhaust
Shading	Solar flux attenuation

Trade secret: Flow coefficients, derating curves, spatial influence functions, and heatmap blending rules are **proprietary** and excluded from this document.

Performance Target

Real-time simulation updates target **sub-50 ms** latency for interactive design workflows. Optimisation techniques are not disclosed.

Continue to *05-Virtual-Experience*

Virtual Experience

Capability Summary

The **Virtual Experience** layer delivers a premium No-Form 3D design environment. Users manipulate greenhouse geometry, crop layout, and climate equipment directly in the viewport while receiving immediate visual and physics feedback.

Experience Modules

Module	User-Facing Capability
Parametric greenhouse	Real-time structural geometry from dimensions
Crop field	Instanced foliage grid scaled to crop profile
Equipment visualisation	Fans, pads, ducts, heaters in 3D context
Climate heatmap	Temperature and VPD floor overlay
Design controls	Sliders and gizmos — no traditional forms
Metrics HUD	Live OPEX, energy, and agronomic readouts

Black-Box Interface

```
/** 3D experience controller – rendering internals are proprietary */
interface VirtualExperience {
  render(design: GrowaDesignContext, snapshot: TwinSnapshot): void;
  setInteractionMode(mode: "orbit" | "scale" | "translate"): void;
  setHeatmapMode(mode: "temperature" | "vpd" | "off"): void;
}
```

Shader & Rendering (Abstracted)

Custom GPU shaders translate simulation heatmap data into colour-mapped floor overlays. Vertex and fragment logic, colour ramps, and equipment-aware spatial blending are **trade secrets** and are not reproduced here.

Internationalisation

Full UI localisation in **English, Italian, Spanish, and French** across all experience modules.

GAIA Intelligence

Home · 05-Virtual-Experience · 07-Data-Governance

Capability Summary

GAIA is Growa's proprietary AI copilot layer for GreenhouseOS. It provides natural-language climate guidance, setpoint recommendations, and design optimisation suggestions grounded in the current virtual twin state.

Design Principles

Principle	Description
Growa-branded	Users interact with GAIA — never underlying providers
Context-aware	Recommendations reflect live design and simulation state
Vendor-aware	Output compatible with industrial climate computers
Resilient	Deterministic local fallback when cloud AI is unavailable

Black-Box Interface

```
/** GAIA copilot – prompt engineering and routing are proprietary */
interface GAIACopilot {
  chat(context: GrowaDesignContext, message: string): GuidanceResponse;
  optimize(context: GrowaDesignContext): SetpointRecommendation[];
  listCapabilities(): ProviderStatus[];
}

interface GuidanceResponse {
  narrative: string;
  recommendations?: SetpointRecommendation[];
}
```

Security Posture

- AI credentials are **server-side only** — never exposed to clients
- Chat context contains simulation parameters only — no user PII beyond optional design name
- System prompts and routing logic are **trade secrets**

Local Optimizer Fallback

When external AI is unavailable, a **deterministic rule engine** produces agronomically sound setpoints. The rule set, thresholds, and vendor tag mappings are proprietary.

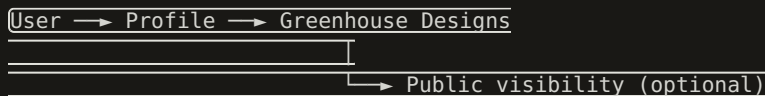
Data Governance

Home · 06-GAIA-Intelligence · 08-Industrial-Bridge

Capability Summary

The **Data Governance** layer manages user identity, greenhouse design persistence, and access control. All data access is enforced at the database level through row-level security policies.

Entity Model (Abstracted)



Entity	Stored Attributes
Profile	Name, company, language preference, AI preferences
Greenhouse design	Geometry, covering, crop, location, metadata

Black-Box Interface

```
/** Data governance – connection details are never documented */
interface DesignRepository {
    list(userId: UserId): GreenhouseDesign[];
    save(userId: UserId, design: GreenhouseDesign): DesignId;
    remove(userId: UserId, designId: DesignId): void;
}
```

Access Control

Policy	Rule
Profile access	User owns their profile
Design access	User owns their designs
Public designs	Readable by all authenticated users

Security Exclusions

The following are **never** included in this vault:

- Database connection strings
- Service role keys
- JWT signing secrets
- Production hostnames

Continue to *08-Industrial-Bridge*

Industrial Bridge

Capability Summary

The **Industrial Bridge** module translates virtual twin outputs into climate-control setpoint bundles compatible with leading greenhouse automation vendors.

Supported Vendor Formats

Vendor	Export Type
Priva	JSON setpoint payload
Ridder	JSON climate rules
Hoogendoorn	JSON process computer setpoints

Export Content (Abstracted)

Category	Examples
Temperature	Day/night targets, ramp rates
Humidity	RH targets per growth stage
Ventilation	Min/max positions, CO ₂ overrides
Screens	Thermal screen conditions
Heating / cooling	Enable thresholds

Black-Box Interface

```
/** Industrial bridge – vendor tag mappings are proprietary */
interface IndustrialBridge {
  export(
    context: GrowaDesignContext,
    snapshot: TwinSnapshot,
    vendor: "priva" | "ridder" | "hoogendoorn"
  ): VendorSetpointBundle;
}
```

Trade secret: Vendor-specific tag prefixes, unit conversions, and crop-stage mapping tables are not disclosed.

Dashboard Metrics

The platform surfaces design-phase estimates for OPEX, energy intensity, and carbon footprint. These are engineering projections, not operational billing data.

Continue to [09-Interface-Contracts](#)

Interface Contracts

[Home](#) · [08-Industrial-Bridge](#) · [10-Trade-Secret-Policy](#)

Public API Surface (Sanitized)

All endpoints are versioned under `/api/v1`. Production origins are environment-configured and not documented here.

Health

```
GET /health
→ { "status": "healthy", "version": "5.0.0" }
```

Simulation

```
POST /api/v1/simulation/run
```

Request shape (abstracted):

```
{
  "climate": { "latitude deg": 0.0, "longitude deg": 0.0, "...": "..." },
  "covering": { "type": "glass", "transmittance": 0.85 },
  "crop type": "tomato",
  "growth stage": "mid season"
}
```

Response shape (abstracted):

```
{
  "et0": { "et0 mm day": 0.0 },
  "agronomic": { "vpd kpa": 0.0, "dli mol m2 day": 0.0 }
}
```

Example values are placeholders. No production coordinates or customer data are used.

Real-Time Stream

```
WS /ws/simulation
```

```
Client → { "event": "UPDATE SIMULATION", "data": { ... } }
Server → { "event": "SIMULATION RESULTS", "data": { ... } }
```

AI · Data · Export

Endpoint	Purpose
GET /api/v1/ai/providers	Provider availability
POST /api/v1/ai/chat	GAIA conversation
GET /api/v1/greenhouses	List saved designs
POST /api/v1/greenhouses	Save design
POST /api/v1/export/climate-computer	Vendor export

Continue to [10-Trade-Secret-Policy](#)

Trade Secret Policy

Home · 00-Cover · 12-IP-Declaration

Protection Boundary

This vault documents the **observable and contractual** surface of the Growa platform. Everything below the interface line is protected.

INCLUDED Interfaces · outcomes ·
capabilities · schema shape
EXCLUDED Algorithms · coefficients ·
prompts · optimisations

Never Disclosed

Category	Examples
Credentials	API keys, tokens, database passwords
Infrastructure	Production URLs, IPs, CDN rules
Algorithms	Solver logic, calibration tables, heatmap blending
AI internals	System prompts, routing rules, fallback thresholds
Business data	Customer records, pricing, unreleased roadmap
Source code	Full implementations, proprietary utilities

Safe Disclosure Checklist

Before sharing this document, confirm:

- ☐ No environment variable **values** appear
- ☐ Code snippets are **interface-only** abstractions
- ☐ No file paths to proprietary algorithm modules
- ☐ No numerical coefficients or tuning parameters
- ☐ Example data uses generic placeholders

Incident Response

If secrets are inadvertently exposed:

- Rotate all affected credentials immediately

2. Regenerate this document
3. Revoke prior distributions
4. Audit service access logs

Continue to [11-Capability-Registry](#)

Capability Registry

Platform Scale (Non-Secret)

Metric	Value
Source modules	~113 files
Lines of code	~15,200
Languages	Python, TypeScript
Locales	4
Milestones delivered	6

Capability Map

Domain	Capabilities	Disclosure Level
Agronomic Intelligence	ET _o , VPD, DLI, stress indices	Outputs only
Climate Twin	Energy balance, equipment, heatmaps	Capabilities only
Virtual Experience	3D viewport, shaders, No-Form UX	Behaviour only
GAIA Intelligence	Chat, optimise, fallback	Interface only
Data Governance	Auth, CRUD, RLS	Schema shape only
Industrial Bridge	Priva, Ridder, Hoogendoorn export	Format names only

Module Categories (Abstracted)

Platform Experience (3D · HUD · Controls · i18n) Orchestration (API · WebSocket · Validation) Agronomic [BLACK BOX] Climate Twin [BLACK BOX] GAIA [BLACK BOX] Data (Auth · Persistence · RLS) Export [BLACK BOX – vendor mappings]

Individual source file paths and internal module names are intentionally omitted to prevent reverse-engineering aids.

IP Declaration

Ownership

Field	Value
Company	Growa
Product	GreenhouseOS 5.0
Copyright	© 2026 Growa / GreenhouseOS Contributors
License	Business Source License 1.1
Change date	2029-08-03 → Apache 2.0

Originality Claims

The following represent **original creative and technical work** by Growa:

- Unified Virtual Twin** — Integration of agronomic physics, real-time climate simulation, 3D design, AI guidance, and industrial export in a single browser platform.
- No-Form Design Paradigm** — Direct in-viewport manipulation replacing traditional configuration forms.
- Client–Server Twin Parity** — Synchronised preview and server-validated simulation for sub-50 ms interactive workflows.
- Equipment-Aware Spatial Visualisation** — GPU heatmap system influenced by climate-equipment layout (proprietary blending).
- GAIA Intelligence Layer** — Growa-branded multi-provider AI with deterministic agronomic fallback.
- Industrial Bridge** — Vendor-specific climate-computer export from virtual twin state.

Third-Party Components

Open-source libraries (FastAPI, React, Three.js, Zustand, Supabase client, Tailwind CSS) are used under their respective licenses. Application architecture, physics integration, and proprietary solvers are Growa intellectual property.

Certification

This document was compiled as a **black-box technical annex** for intellectual property registration. It contains no credentials, production infrastructure details, proprietary algorithms, or full source code, in accordance with [10-Trade-Secret-Policy](#).